Research on Security Issues of Artificial Intelligence

Paxuan Fan
Shanghai Urban Construction Vocational College
Shanghai, China
fanpaxuan@163.com

Gaofeng Xu*
Shanghai Urban Construction Vocational College
Shanghai, China
pc1616@126.com

Abstract

In this scholarly exploration, the complexities and vulnerabilities associated with artificial intelligence utilization in variegated domains are scrutinized. Evident from widespread AI deployment is a significant increase in its inherent security risks. Investigated within this manuscript are nuanced facets of data confidentiality compromises, algorithmic integrity concerns, privacy-related dilemmas as well as ethical quandaries borne out of moral conundrums intrinsically linked to intelligent system implementations. Moreover, corresponding defense mechanisms tailored to mitigate these threats find propositions herein—inclusive through lenses technological, operational governance, alongside legislative oversight frameworks—analyzed in detail for strategic enhancement and fortification against emergent digital hazards.

CCS Concepts

• **Security and privacy** → Software and application security; Domain-specific security and privacy architectures.

Keywords

Artificial Intelligence, Security, Data Security, Algorithm Security, Ethical and Moral

ACM Reference Format:
Paxuan Fan and Gaofeng Xu. 2025. Research on Security Issues of Artificial Intelligence. In *2025 2nd International Conference on Image Processing, Intelligent Control and Computer Engineering (IPICE 2025), July 25–27, 2025, Zhengzhou, China.* ACM, New York, NY, USA, 5 pages. https://doi.org/10.1145/3768184.3768201

1 Introduction

Attracting global attention in recent years, the impressive prowess of language processing capabilities and extensive variety of application scenarios pertaining to generative forms of artificial intelligence exemplified by ChatGPT have.

Observed from recent trends, the security vulnerabilities inherent within artificial intelligence systems are manifesting with heightened clarity. Undertaking scholarly inquiries of a comprehensive nature concerning the fortification measures requisite for artificial intelligence implementations emerges as an area imbued with profound practical import. Apprehending latent susceptibilities and proffering strategies fortified with efficacy in safeguarding against breaches reflects not merely on the protection of user rights deemed legitimate but also aims to engender public confidence towards this evolving realm of technology [1-3]. Concomitantly contributing does this to the robust progression and enduring viability of industries centered around artificial intelligence advancements.

Comprehensively and aiming for depths previously unachieved, this inquiry scrutinizes the protective measures pertinent to artificial intelligence implementations. Various research methodologies are applied in a holistic approach, wherein dimensions of multiple nature inform each phase of analysis. From such a methodological framework, birthed are strategies characterized not only by innovation but feasible applicability, indicative of strivings to elevate the security discourse to new echelons.

Firstly, to commence with, the methodology of literature exploration is employed. Through the examination and comprehension of antecedent research paradigms, methodological approaches, and prevailing theoretical perspectives within the domain of artificial intelligence security, elucidation emerges concerning contemporary critical challenges and pressing issues. Insight from this scrutiny instills a robust epistemological basis alongside innovative conceptual paradigms for this scholastic inquiry.

In addition, interdisciplinary research methodologies were further utilized within this inquiry [4-6]. Encompassing the domain of artificial intelligence security, there coexist disparate fields including computer sciences, legal frameworks, ethical considerations, mathematical formulations and sociological perspectives. It is apparent from such instances that solitary disciplinary approaches confront substantial challenges when endeavoring to address multifaceted issues of security in a comprehensive manner. Therefore, this study comprehensively applies theories and methods from various disciplines to analyze artificial intelligence security issues from multiple perspectives such as technology, law, ethics, and society.

An exploration marked by considerable inventiveness is nestled within the primary contours of this investigation, manifested chiefly through a research lens enveloping multiple dimensions in tandem with an encompassing array of security stratagems. Of particular significance is the integration of variables legalistic, ethical, societal, amongst others multifarious in nature, coalescing into the field of scrutiny. Instances reflected therein suggest that during the formulation of stratified approaches toward securing protection, it becomes paramount to acknowledge the synchronistic interplay among jurisprudence, technology, ethical reasoning, and societal governance constructs, thereby engendering a systemic scaffold enshrining multi-tiered, exhaustive protective layers surrounding artificial intelligence mechanisms. It can be seen from technical strata referenced [7-8] a proactive recommendation surfaces: it prompts intensification of efforts directed towards investigational pursuits alongside developmental undertakings apropos artificial

*Corresponding author.

IPICE 2025, Zhengzhou, China

ACM ISBN 979-8-4007-2111-3/2025/07
https://doi.org/10.1145/3768184.3768201

intelligence technological safeguards. At the legal level, it is recommended to improve relevant laws and regulations, clarify the definition of responsibilities and legal norms in the application of artificial intelligence [9-10]; At the ethical level, advocating for the development of ethical guidelines for artificial intelligence to guide technology developers and users in adhering to ethical standards.

2 Insight into Artificial Intelligence Security Issues

2.1 Data security risks

Data leakage risk: The processes of aggregation, retention, and dissemination are the conduits through which data traverses. Consequences ensue upon the unfortunate eventuation of a breach in data integrity—consequences manifesting as grave damage to individual privacies and entitlements, exacerbated further by monumental repute-wise and fiscal detriments inflicted upon affiliated enterprises and establishments. Evidences from this phenomenon substantiate that not only did Equifax endure substantial liabilities alongside juridical proceedings,but an egregious rift in consumer confidence materialized vis-à-vis both the corporation itself and the broader industry's assurances concerning data securitization.

Data transmission risk: The openness of network transmission makes data vulnerable to theft or tampering. In cloud computing environments, data needs to be transmitted between different servers. If encryption measures are not taken during transmission, data may be intercepted by hackers.

Data tampering risk: Trained on datasets constrained to the temporal boundary of October 2023, one may discern that models constructed upon erroneous data provocations instigated by malevolent entities possess a propensity for assimilating fallacious information. This phenomenon often precipitates an egregious diminution in decisional veracity and portends dire ramifications across spheres of paramount importance.

In the field of autonomous driving, training data is crucial for the accuracy and safety of autonomous driving models. If an attacker maliciously tampers with training data, the decision-making system of autonomous vehicle may receive wrong information, thus making wrong driving decisions, causing traffic accidents and endangering the lives of passengers and pedestrians.

In the sphere of finance, an arena particularly susceptible to data manipulation hazards one might argue is encountered. The employment within this sector of artificial intelligence, notably in appraising risks financially and in the formulation of decisions regarding trades—this phenomenon observes a proliferation increasingly notable. Tamper with training datasets once they are, erroneous judgments regarding fiscal vulnerabilities by financial institutions could arise, leading them towards flawed investment decision-making processes; it can be seen that eminent economic detriments may then inevitably ensue.

The issue of data security is essentially a dual product of technical vulnerabilities and lack of legal supervision. From a computer science perspective, the application of data encryption algorithms such as AES can resist transport layer attacks, but legal compliance standards for encryption technology need to be clearly defined (such as GDPR's mandatory requirements for data encryption). Sociological research shows that public concerns about data abuse can reduce technological acceptance, and trust needs to be established through ethical review mechanisms such as the transparency provisions of the EU AI Act. For example, medical data leakage incidents not only involve technical protection vulnerabilities, but also require tracing the responsible party in accordance with the Personal Information Protection Law, and evaluating the legitimacy of data use through an ethics committee.

2.2 Algorithm Security Vulnerabilities

Resistance sample attack: In the field of image recognition, attackers generate adversarial samples by adding small and imperceptible perturbations to the original image, causing the image recognition model to misjudge. Taking the visual recognition system of autonomous vehicle as an example, its normal operation depends on the accurate recognition of various signs, obstacles and other vehicles on the road.

To the phenomenon under consideration, algorithm bias can be attributed to an amalgamation of data prejudices and imperfections within algorithm configurations, exhibiting a proclivity for engendering inequitable outcomes in decision-making procedures undertaken by algorithms. Observable in recruitment, firms implement AI algorithms designed for culling through resumes and assessing potential talents. It is apparent from instances that due to historic tendencies embedded within training datasets—favoritism towards specific genders, ethnicities, or scholastic credentials—algorithms may perpetuate and exacerbate such preconceptions, culminating in unjust appraisals of candidates pursuing employment opportunities.

2.3 Privacy Protection Dilemma

The risk of violating user privacy exists in many fields. In the field of smart homes, devices collect a large amount of user data during operation, and there are many risks of violating user privacy in this process. As a typical representative of smart homes, smart speakers continuously collect user voice commands, daily conversations, and other data during voice interaction. If its security measures are not in place, hackers can easily invade the camera system, obtain real-time footage captured by the camera, and completely expose the user's home life to the gaze of others.

The application of privacy protection technology in artificial intelligence faces many challenges. Differential privacy, as a commonly used privacy protection technique, achieves privacy protection by adding noise to the data. When using differential privacy technology, a difficult balance needs to be struck between data accuracy and privacy protection. Homomorphic encryption has high computational complexity and consumes a significant amount of computing resources and time when performing complex computational tasks.

Beyond the realms of differential privacy and homomorphic encryption methodologies, it becomes apparent through various instances that additional privacy safeguarding technologies confront significant challenges within the domain of artificial intelligence's applications. High communication costs to consider; this is an issue for secure multi-party computation technology—subject-verb inversion here required—which must navigate complications when executing collaborative data computations amongst manifold participants: low computational efficiency and protocol security being pivotal concerns. It is observed that federated learning methods enable joint modeling devoid of raw data exchange—the main concern remains ensuring the steadfast protection and secrecy of data from each involved party—a pressing matter yet to find resolution determinate.

2.4 Ethical and Moral Disputes

There is much controversy over the attribution of responsibility for artificial intelligence decision-making. Taking the autonomous vehicle accident as an example, when the AI system makes decisions that lead to adverse consequences, responsibility determination becomes extremely complex. From a technical perspective, artificial intelligence systems make decisions based on pre-set algorithms and models, and the design and training process of algorithms may have flaws, leading to decision-making errors. Enterprises developing auto drive system may think that they have designed and tested algorithms according to industry standards and specifications, but due to the complexity and uncertainty of actual road conditions, accidents cannot be completely avoided.

Lack of moral and ethical standards. In the medical field, when using artificial intelligence assisted diagnostic systems, doctors may overly rely on the system's diagnostic results and neglect their professional judgment. Some complex disease symptoms may have multiple possibilities, and artificial intelligence systems may not fully consider all factors. If doctors only treat based on the diagnostic results of the system, it may lead to misdiagnosis or mistreatment, affecting the health and safety of patients. In the military field, autonomous weapon systems are one of the important applications of artificial intelligence, which can autonomously identify targets and launch attacks without human intervention. However, the use of this autonomous weapon system has sparked serious ethical controversy as it may lower the threshold for war and make it easier for decision-makers to launch attacks without worrying about casualties.

3 Exploration of Security Strategies

3.1 Strengthen data security management

Indispensable for safeguarding data is encryption, data protection's essential mechanism. Transforming plaintext into ciphertext via dedicated algorithms it achieves; thus recoverability by only decryption-key-owning authorized entities is ensured. Widely implemented is the symmetric encryption algorithm, exemplified superbly by AES—Advanced Encryption Standard. In opposition, asymmetric encryption employs dual keys—a public-private dyad facilitating encryption-decryption processes—with RSA standing as a quintessential model therein. Abstract encryption or hashing functions predicate integrity verification in data contexts greatly. It emerges distinctly observable from prevalent vulnerabilities that MD5, having manifestly decayed security robustness, now cedes place to enhanced hashes like SHA-256 inherently more secure. Crucial becomes the effectiveness of encryption through particular emphasis on adept key stewardship management. Embedded rigorously within this spectrum lies Hierarchical Key Architectures conjoined with Hardware Security Modules (HSM), ensuring comprehensive lifecycle guardianship over pivotal cryptographic assets tactfully administered.

Data desensitization is the process of converting, masking, or replacing sensitive data without affecting its value. Substitution, masking, and encryption are common methods for data anonymization. In artificial intelligence data processing, data encryption and anonymization technologies should be comprehensively applied based on the characteristics of the data and the requirements of the application scenario, to construct a multi-level data security protection system.

The Role Based Access Control (RBAC) model is a widely used data access control model that introduces the concept of "roles" between users and permissions. By assigning permissions to roles and then assigning roles to users, decoupling between users and permissions is achieved, greatly simplifying permission management work. The Attribute Based Access Control (ABAC) model is a more flexible authorization model that can dynamically calculate whether one or a set of attributes meet certain conditions for authorization judgment.

In the realm of medical artificial-intelligencious applications, one observes dynamically authorized access control to patient medical record data founded upon elements such as profession-related domain of the physician, exigency characteristic of the patient's situational condition, and temporal aspect concerning access. Notably seen in instances where normal working hours find the primary care provider interacting with said records, authorization by system mechanism occurs automatically. Accessions into patients' medical records during non-urgent instances by physicians from disparate departments necessitate rigorous procedural approval, however. The ABAC schematization, it is inferred from practice, demonstrates adaptability conducive to complex transformational business paradigms alongside demanding security does not circumvent finesse-grained controldoms but imposes implementation complexities requiring meticulous definitional management across diverse attribute spectrums.

In order to prevent data abuse and leakage, strict permission management strategies should also be developed, following the principles of minimum permission and minimum leakage. The principle of minimum privilege refers to following the principle of minimizing privilege when assigning privileges to subjects, limiting the implementation of authorized operations to the maximum extent possible, and avoiding unexpected events, errors, and dangers caused by unauthorized subjects.

Establishing a comprehensive permission audit mechanism is also an important component of data access control. By recording and analyzing user access behavior, potential security risks can be identified in a timely manner, such as abnormally frequent access, unauthorized access attempts, etc. The permission audit mechanism can also provide a basis for traceability and responsibility determination after an event. Once a data security incident occurs, relevant access operations and responsible persons can be traced through audit logs.

3.2 Enhancing Algorithm Security

Improving algorithm security requires enhancing algorithm robustness. Adversarial training is an important technical means to improve the robustness of algorithms. By introducing adversarial samples during the training process, the model learns resistance to these adversarial samples, thereby improving its stability and accuracy in the face of various interferences. Model fusion is also an effective method to enhance the robustness of algorithms.

Visualization technology is an intuitive and interpretable method that presents the internal state and decision-making process of algorithms in a graphical manner, helping users understand the behavior of algorithms more intuitively.

Rule extraction is another important interpretability algorithm that aims to extract easily understandable rules from complex models to explain their decisions. In the decision tree model, it already has good interpretability, and users can directly read decision rules from the decision tree. For more complex deep learning models, decision rules can also be obtained through rule extraction algorithms. For deep learning models used for disease diagnosis, rule extraction algorithms can be used to transform the model's processing of input data such as patient symptoms and examination results into rules such as' if symptom a and examination result B occur simultaneously, diagnose as disease C '. Doctors can use these rules to understand the diagnostic basis of the model and assess the validity of the diagnostic results.

3.3 Improve privacy protection mechanisms

In the realm of federated learning, cooperative modeling amidst numerous entities is attainable whilst abstaining from the transference of unprocessed data; therein lies its ability to thoroughly extract the inherent worth within datasets even as it safeguards their privacy. As observed in medical enterprises, considerable volumes of patient medical records exist dispersed amongst diverse healthcare establishments—yet impediments arise from privacy legislation and regulatory frameworks, rendering such data challenging for direct interchange or amalgamation. Medical institutions, through said technological advancement, maintain autonomy over original datasets by securing them locally while solely disseminating encrypted model parameters or intermediary computational outcomes. On the server end are these disparate elements subsequently coalesced—model aggregation achieved—and updates executed accordingly.

Multi party secure computation is another important privacy protection technology that allows multiple parties to jointly calculate the objective function without leaking their own private data. Throughout the entire computation process, the data of all parties is always kept encrypted and will not be leaked to other participants, ensuring the privacy and security of the data. Multi party secure computation can also be applied to the training of machine learning models, achieving joint modeling under data privacy protection and improving model performance and accuracy.

Differential privacy is also a commonly used privacy protection technique, which achieves privacy protection by adding noise to the data. In data analysis and statistical scenarios, differential privacy techniques can be used to protect the privacy of personal data while obtaining statistical features of the overall data. When conducting population census data analysis, in order to protect the personal information of each census subject and obtain accurate demographic results, such as population age distribution, gender ratio, etc., an appropriate amount of noise can be added to the statistical data. By adjusting the noise intensity reasonably, it is possible to ensure the accuracy of statistical results while maximizing the protection of personal data privacy. Differential privacy technology is also applied in artificial intelligence model training, by adding noise to the training data to prevent attackers from inferring the original data through the model, thus protecting data privacy.

3.4 Building an ethical and moral framework

In the elaboration of ethical-moral frameworks governing artificial intelligence, there emerges a necessity for adherence to principles wherein human interests and well-being are prioritized. The overarching objective should ensure that artificial intelligence, in its development and usage, aligns with humanity's long-term benefit continuum. Manifested within the paradigmatic content of these standards — divergent dimensions such as privacy protection, data security, algorithmic fairness, alongside accountability can be observed from varied perspectives.

When focus diverges towards privacy guardianship, prescriptive elements must articulate mandatory compliance by AI systems with stringent privacy doctrines during the aggregation, deployment, and retention of user-centric data. Explicit user consent acquisition juxtaposed against effective encryption measures safeguarding data confidentiality is thus indispensable.

Attention shifts thereafter to the imperative domain of informational security; an evident obligation surfaces for artificial intellect constructs to embody robust mechanisms permitting comprehensive data redundancy and restitution coalescence - thereby averting potential loss or corruption scenarios. Furthermore, the establishment of rigorous access oversight procedures delineates a critical aspect geared towards thwarting illicit exfiltration and exploitation of information repositories.

Algorithm fairness introduces another echelon requiring developers' meticulous attentiveness towards inclusivity therein. Compulsory scrutiny pertains to datasets' diversity and representational paradigms - preventive of biasically-skewed computational outputs. Rendering visible the decision pathways administered by algorithms serves an additional tier—it facilitates comprehension empowerment amongst stakeholders—thus ensuring equitability and equity reaffirmations audience-directed.

Responsibility assignment constitutes one final thematic axis demanding detailed articulation through guidelines. Observably specified are roles along with consequential accountability transferences attendant upon malfunction cases or negative outcome emergences associated with AI systems— this serving as a deterrent against blame transference amidst involved entities.

4 Conclusion and Prospect

This study mainly focuses on the security of artificial intelligence applications, deeply analyzes their current situation and problems, and proposes comprehensive security strategies. Through multi-dimensional research methods, a series of valuable achievements have been made.

A thorough understanding of security issues is a key part of this study. Clearly identify data security risks, such as the common risk of data leakage in data collection, storage, and transmission, and the Equifax data breach incident is a typical case; The threat of data tampering may lead to artificial intelligence models learning incorrect information, affecting decision accuracy, and potentially causing serious consequences in the fields of autonomous driving and finance. In terms of algorithm security vulnerabilities, adversarial sample attacks can lead to incorrect judgments in image recognition models; The problem of algorithm bias arises from data bias and algorithm design flaws, which may lead to unfair decisions in the recruitment and judicial fields, damaging social fairness and justice. The dilemma of privacy protection is reflected in the infringement of user privacy, and there are many privacy risks in the data collection and use process of smart homes and smart wearable devices;

An intricate security paradigm is proposed herein, derived from meticulous examination of multifaceted security challenges. Tools and methods pertaining to data security management, encryption and anonymization—integral elements like the AES cryptographic approach, hashing techniques, substitutionary procedures, alongside masking strategies represent avenues ensuring data's inviolability and integrity. Access controls over data necessitate imposition of stringent measures; role-based (RBAC) and attribute-based access control (ABAC) models ensure adherence to principles of minimal privilege coupled with least leakage parameters, where audit enhancements on permissions deter potential abuses and leak vulnerabilities.

Algorithmic robustness sees elevation through adversarial methodology training combined with model amalgamation paradigms, emerging in resistance enhancement against antagonistic samples—and thereby projecting a fortified generalizing capability across integrated models. Algorithm interpretative depth demands augmentation via visualization paradigms aligned with rule-extraction methodologies, rendering comprehensible transparency within decision-making apparatuses which aids anomaly identification leading towards preemptive rectifications.

Enhancement of privacy protection systems incorporates technologies such as federated learning, secure multiparty computation along with differentially private mechanisms. From this manifests clearly articulated stipulations surrounding privacy protocol definitions aligning regulatory frameworks with system implementation. This study comprehensively reveals the security issues in the application of artificial intelligence, proposes a systematic security strategy, provides theoretical support and practical guidance for the safe development of artificial intelligence, and helps promote the safe and reliable application of artificial intelligence technology in various fields. Future research can further explore adaptive security policy generation algorithms to achieve automatic derivation and optimization of security protection strategies for artificial intelligence systems during technological evolution, promoting the formation of a positive cycle of "technological development security evolution".

References

[1] Goodfellow I J, Shlens J, Szegedy C. Explaining and harnessing adversarial examples [J]. arXiv preprint arXiv:1412.6572, 2014.

[2] Zhao Xue, Zheng Qinghua, Shen Chao, etc. A Review of Artificial Intelligence Security Research [J]. Journal of Software, 2020, 31 (10): 2941-2974.

[3] Liu Kai, Wang Feiyue. Artificial Intelligence Security and Rule of Law: Issues, Challenges, and Prospects [J]. Journal of Automation, 2020, 46 (4): 647-658.

[4] Zhang Xinbao. From Privacy to Personal Information: Theoretical and Institutional Arrangements for Rebalance of Interests [J]. Chinese Journal of Law, 2015 (3): 38-59

[5] Cai Zixing, Xu Guangyou. Artificial Intelligence and Its Applications [M]. 5th edition. Beijing: Tsinghua University Press, 2016

[6] Bogoviz A V , Alekseev A N , Lobova S V ,et al.A new educational model for training digital personnel and intelligent machines: standardisation vs. flexibility[J]. 2024.

[7] Manu H N , Onyame E A , Mensah E A ,et al.Knowledge Modeling and Institutional Memory at the University of Cape Coast: Examining Technology as a Mediator and Leadership Styles as a Moderator in Enhancing Administrative Efficiency[J].Intelligent Information Management, 2025, 17(1):30.DOI:10.4236/iim.2025.171001.

[8] Guo X .Research on Innovation and Entrepreneurship Talent Cultivation Mode Innovation with Artificial Intelligence Technical Support[J].Applied Mathematics and Nonlinear Sciences, 2024, 9(1).DOI:10.2478/amns-2024-3039.

[9] The full text of the General Data Protection Regulation (GDPR) of the European Union [EB/OL]. (April 27, 2016) [March 10, 2024] https://eur-lex.europa.eu/eli/reg/2016/679/oj.

[10] Personal Information Protection Law of the People's Republic of China [EB/OL]. (2021-08-20) [2022-03-10] http://www.npc.gov.cn/npc/c30834/202108/677c0c1c60724d0797546c1c8079e595.shtml